Hands-On AI: Automate Data Analytics and Reporting with n8n

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LEARNING

Installing n8n: Cloud vs. Self-hosted

When considering whether or not to self-host, the most important question to consider is “how customized does your experience need to be?” If you have a higher than typical need for security, data processing, or scalability, then self-hosting will probably be the best fit for you. Otherwise, for optimal user experience, the n8n cloud is a great option.

Category	Self-Hosted n8n	n8n Cloud	Winner
Data Connectivity and Security	Direct access to internal databases (SQL Server, Postgres) without whitelisting IPs. Sensitive data stays on your infrastructure - critical for highly regulated industries.	Best for SaaS tools (Salesforce, Slack, Google Sheets). Internal database access requires firewall changes or VPN tunnels, which IT often blocks.	✓ Self-Hosted
Python and Data Processing	Custom Docker images with any Python libraries (pandas, numpy, scikit-learn). Full hardware control. Can scale to 32GB RAM for heavier compute intensive jobs.	Standard environment for lightweight logic. Large datasets (>50MB CSVs) can crash or time out due to memory limits.	✓ Self-Hosted
Cost and Scalability	Flat server cost (~$20/ month VPS) regardless of executions. Free Community Edition available (lacks SSO and team features).	Pay-per-execution pricing. High-frequency workflows (polling, Slack listeners) can quickly exceed quotas and incur overage fees.	✓ Self-Hosted
Maintenance and IT Friction	You handle Docker updates, backups, server security, and version rollbacks. Downtime is your responsibility.	Zero maintenance—n8n manages upgrades, security patches, scaling, and uptime.	✓ Cloud

Collaboration and Teamwork	Free version lacks granular RBAC—colleagues get full admin access. Enterprise license required for team features.	Team-ready with user management on lower-tier plans. Easier to share workflows safely.	✓ Cloud

To get started with self-hosting an n8n instance, here is the n8n official guide: https://docs.n8n.io/hosting/